TAGUCHI'S METHOD, SIX SIGMA, ISO9001

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A 2^3 full factorial array

Experiment	A	B	C	A×B	B×C	C×A	A×B×C
1	-1	-1	-1	+1	+1	+1	-1
2	-1	-1	+1	+1	-1	-1	+1
3	-1	+1	-1	-1	-1	+1	+1
4	-1	+1	+1	-1	+1	-1	-1
5	+1	-1	-1	-1	+1	-1	+1
6	+1	-1	+1	-1	-1	+1	-1
7	+1	+1	-1	+1	-1	-1	-1
8	+1	+1	+1	+1	+1	+1	+1

$$\text{Effect of a factor} = \frac{\sum \text{responses at level 2 of the factor} - \sum \text{responses at level 1 of the factor}}{\text{half the number of experiments}}$$

$$\text{Quadratic effect} = \frac{\sum \text{responses at level 2} - 2\sum \text{responses at level 3} + \sum \text{responses at level 1}}{\text{one third the number of experiments}}$$

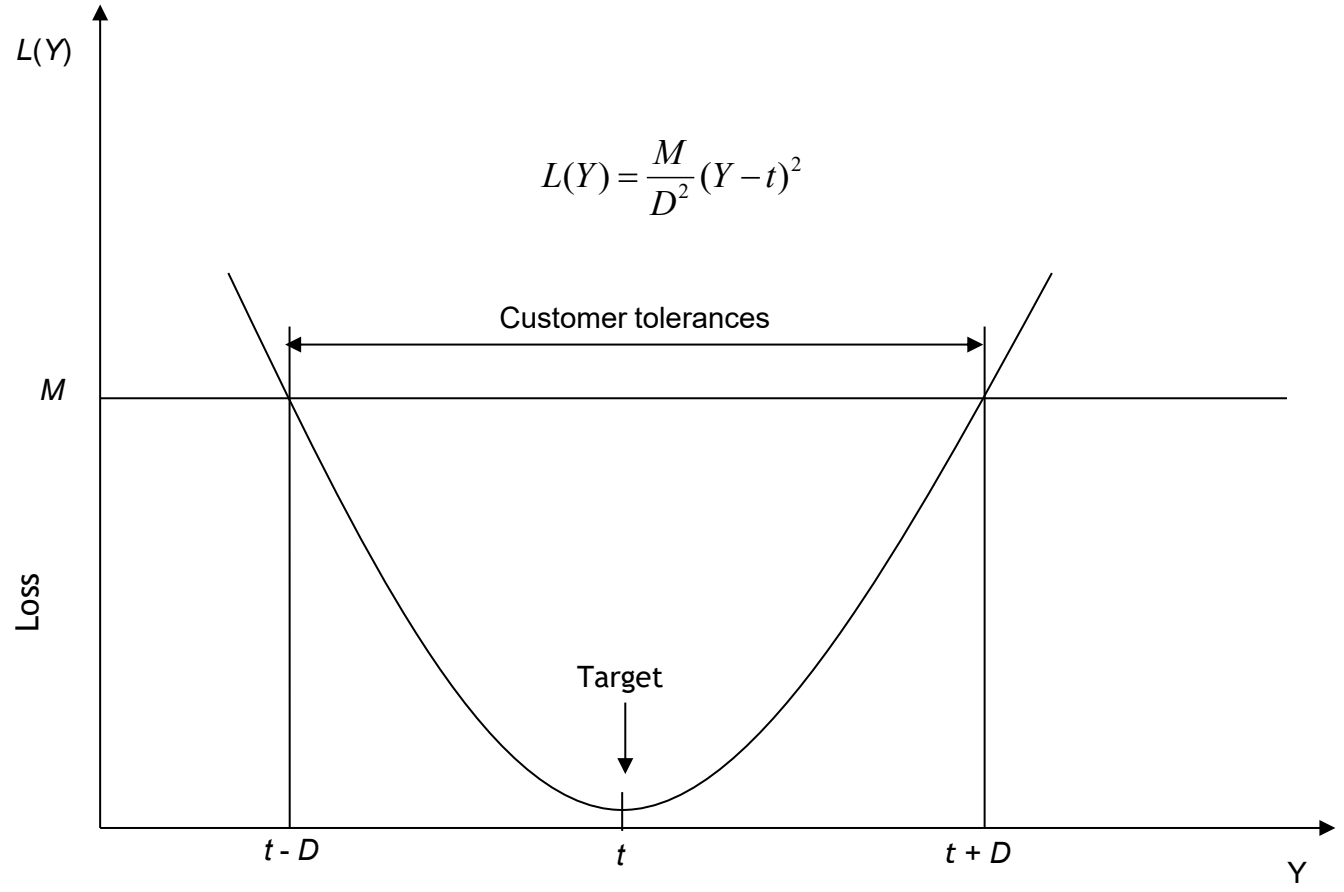
Level 3 is the intermediate level.

An OA_4 (2^3) fractional factorial array

Experiment	A	B	C	A×B	B×C	C×A	A×B×C
2	-1	-1	+1	+1	-1	-1	+1
3	-1	+1	-1	-1	-1	+1	+1
5	+1	-1	-1	-1	+1	-1	+1
8	+1	+1	+1	+1	+1	+1	+1

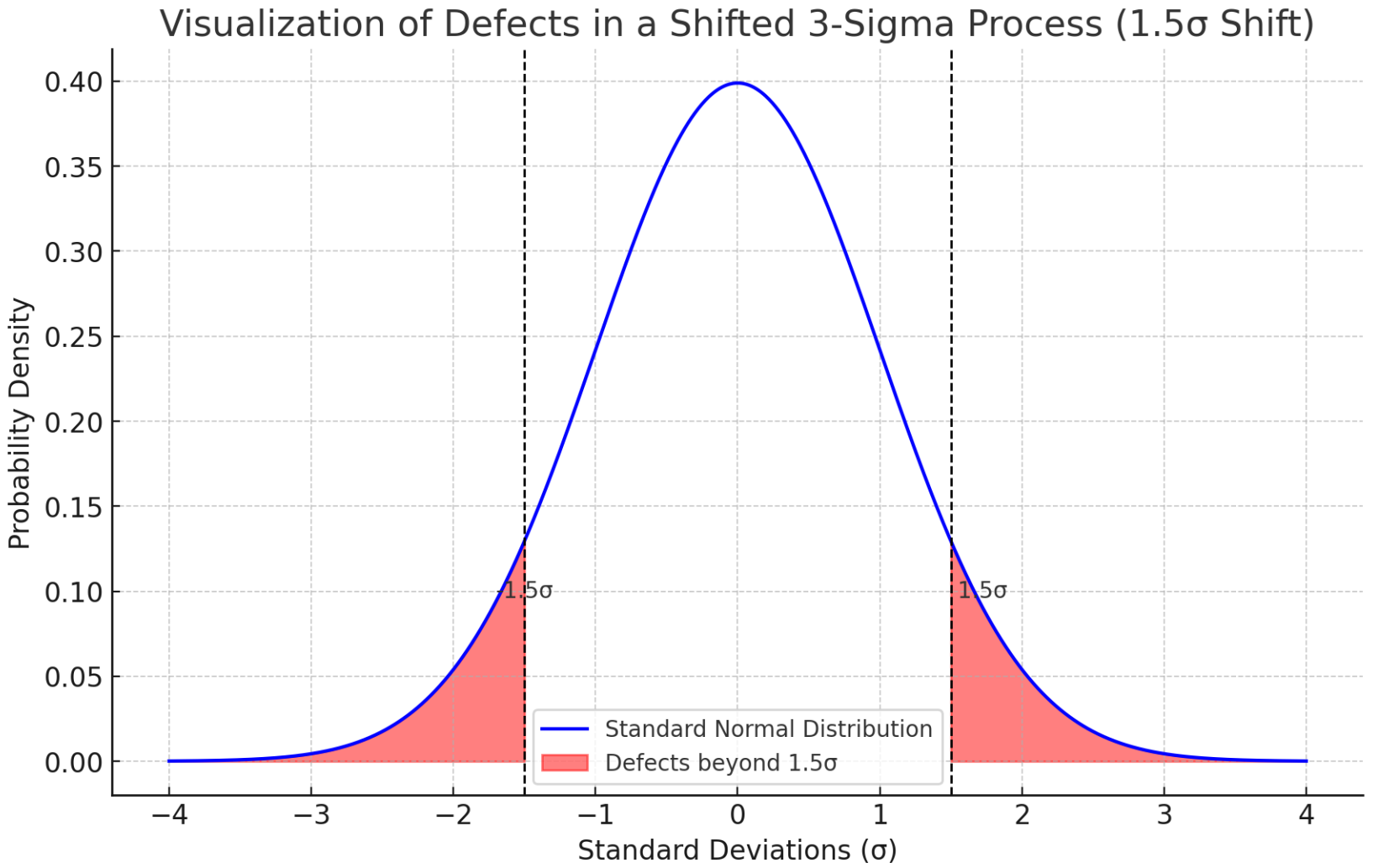
Taguchi's loss function

$$L(Y) = \frac{M}{D^2}(Y - t)^2$$



Six Sigma

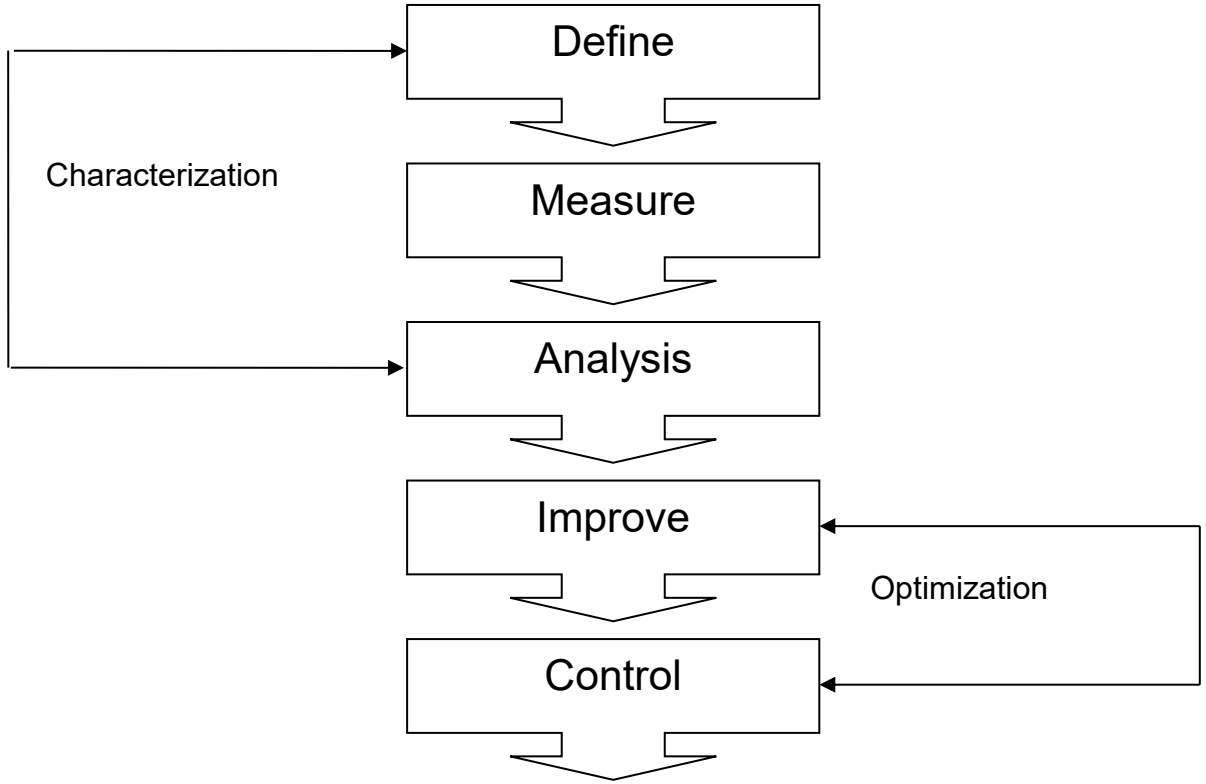
- Six Sigma was developed by Bill Smith and Mikel Harry, both engineers at Motorola in the 1980s.
- Specifically, Smith is often credited as the "father of Six Sigma" for first coining the term and identifying the need to set specific improvement goals based on defect measurement.
- It is a high performance, data driven approach for analyzing the root causes of business processes/ problems and solving them. It links Customers, Process improvements and financial results.
- Sigma is a Greek alphabet and is used in statistics as a measure to denote the standard variation in a process.
- More specifically sigma measures the capability of the process to perform defect free work.
- A defect is anything that results in customer dissatisfaction like defective component, wrong shipment, delayed deliveries. The sigma value indicates how often defects are likely to occur.



Many manufacturing company processes operate at 3 sigma level which translates into approx. 67000 defects per million, while service company processes are often at 1 to 2 sigma level i.e. 690,000 to 308,000 defects per million.

The area under the curve of a normal distribution between -4.5 sigma and +4.5 sigma (or within 4.5 standard deviations of the mean) represents approximately 99.99966% of the total area. This corresponds to a 3.4 defect rate per million opportunities, which is the basis for the Six Sigma methodology.

Six Sigma Methodology



1. **Defects Per Unit (DPU)**

$$\text{DPU} = \frac{\text{Total Defects}}{\text{Total Units Produced}}$$

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2. **Defects Per Million Opportunities (DPMO)**

$$\text{DPMO} = \frac{\text{Total Defects}}{\text{Total Units} \times \text{Opportunities per Unit}} \times 1,000,000$$

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3. **Yield**

$$\text{Yield} = \left(1 - \frac{\text{Defective Units}}{\text{Total Units}}\right) \times 100\%$$

4. **First Pass Yield (FPY)**

$$\text{FPY} = \frac{\text{Units with No Rework}}{\text{Total Units Processed}} \times 100\%$$

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5. **Rolled Throughput Yield (RTY)**

(For multiple steps)

$$\text{RTY} = \prod_{i=1}^{n} \text{FPY}_i$$

where n is the number of process steps.

$$C_p = \frac{\mathrm{USL} - \mathrm{LSL}}{6\sigma}$$

Six sigma Belts

- White Belt: Basic understanding
- Yellow Belt: Team member on projects
- Green Belt: Project leaders, DMAIC methodology
- Black Belt: Full-time improvement experts
- Master Black Belt: Strategy and mentoring

ISO 9001:2015 – Quality Management System (QMS)

- ISO 9001:2015 is an internationally recognized standard for Quality Management Systems (QMS) published by the International Organization for Standardization (ISO).
- It provides a systematic framework to ensure that organizations consistently meet customer and regulatory requirements while improving overall efficiency and performance.

Key Features of ISO 9001:2015

- **Process Approach** – Emphasizes understanding and managing interrelated processes for efficiency.
- **Risk-Based Thinking** – Identifies and mitigates potential risks in quality management.
- **Customer Focus** – Ensures customer satisfaction by consistently delivering quality products/services.
- **Leadership Involvement** – Encourages active participation of top management in quality initiatives.
- **Continuous Improvement** – Fosters a culture of ongoing enhancement of processes.
- **Evidence-Based Decision Making** – Relies on data analysis for effective decision-making.
- **Context of the Organization** – Considers internal and external factors affecting quality.

Benefits of ISO 9001:2015

- Enhances customer satisfaction and trust.
- Improves operational efficiency and reduces errors.
- Increases market credibility and business opportunities.
- Ensures compliance with regulatory and statutory requirements

ISO 9001:2015 is applicable to organizations of all sizes and industries, making it a versatile tool for quality assurance and business excellence.

OSHA: Ensuring Workplace Safety

Full Form: Occupational Safety and Health Administration
Established: 1970 (under the Occupational Safety and Health Act)
Agency of: U.S. Department of Labor

Key Responsibilities of OSHA

- Set and enforce workplace safety and health standards
- Provide training, outreach, education, and assistance
- Inspect workplaces for compliance
- Protect workers from known hazards

What does OSHA contribute?

- Reduces workplace injuries and illnesses
- Promotes safe and healthful working conditions
- Empowers employees to report unsafe practices